

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	COTF-0000000003
Award Program	Crops of the Future Direct Fund
Project Title	Maize Resilient to Transient and Long-term Drought
Grant Start Date	12/31/2019
Grant End Date	6/30/2022
Total Grant Budget	\$1.7 million
Total FFAR Budget	\$1.7 million
Matching Funder(s)	Syngenta, Inari, KWS
Final Report Date	11/30/2022
Project Director/Principal Investigator Information	
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Grantee Organization Information	
Organization Name	University of Wisconsin-Madison

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1. General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Madison, WI; 2nd Congressional District

State College, PA; 12th Congressional District

East Lansing, MI; 8th Congressional District

Graneros, Chile

- 1.2. How many new jobs were created because of this grant funding?
8

- 1.3. How many jobs were maintained because of this grant funding?
1

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.
No

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.
No

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be ‘stand-alone’ and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Drought is a major challenge to agricultural production which underlies US and global food and raw material production. Climate change has increased the frequency and intensity of long-term drought and has also increased variability in climate resulting in greater transient drought stress during the growing season. Mitigation of drought through irrigation is energy-intensive and often accompanied by environmental cost and social justice issues. Breeding cultivars that are tolerant to transient and long-term drought is a critical component of supporting agricultural productivity and resilience in the face of climate change. This research evaluated experimental, biological, and computational approaches to enhance breeding of drought tolerant cultivars. Approximately 3,000 maize hybrids were grown in managed-stress trials across the three years of the study. The well-watered treatment was designed to minimize stress and the drought treatment withheld water across the flowering period with a goal of reducing yield ~30% relative to the well-watered treatment. Performance traits were measured including grain yield, time from flowering to planting and flowering delay due to drought, and plant responses to drought such as leaf-rolling and reduction in overall biomass. Image analysis of plants and ears supported measured traits. Data from the trial was used to evaluate statistical models that could enhance breeding progress and was compared to large-scale public trials that included an array of realized stresses. In addition, the trial was used as a substrate for physiological analysis which led to discovery of a new xylem trait which documented the importance of the plant “plumbing system” in drought resistance. Advanced computational approaches supported the utility of genome-based breeding methods in breeding for drought tolerance. A workshop was held for collaborators on statistical and breeding methods. In sum, this project provided new insights that will support future breeding efforts for drought tolerance and contributed to the training of future plant scientists.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

The key findings of this research will support future breeding efforts and have provided information in use at the University of Wisconsin maize breeding program. The findings will support future breeding decisions in several ways. The project utilized commercially relevant germplasm and the information on germplasm sources and useful alleles can be directly incorporated in the breeding process. Second, the project considered physiological and image-based phenotypic measurements that are not commonly used in maize controlled-stress trials providing insights on the utility of potential tools for breeding programs. Finally, the data sets, integrated data summaries, and analysis pipelines have been shared with the funding collaborators and will become a public resource through direct access to data and publications nearing completion.

Key findings:

1. Elite non-Stiff Stalk inbred parents PHN46 and PHP02 were identified as candidate lines to support high-yield and stress tolerance. In a separate study, PHN46 was also identified as a parent of progenies that had drought and low N tolerance. Variation for drought tolerance was substantial in a multi-parent Stiff Stalk population and alleles for drought tolerance in this heterotic group were identified. Mo44 was identified as a parent with tropical germplasm heritage that is a source of useful alleles for drought tolerance. In summary, the study provided information on germplasm sources that can be used to improve drought tolerance in U.S. breeding programs.
2. Managed drought trials support identification of hybrids, and development of prediction models, that can be used in breeding processes to develop cultivars that will support productivity in a changing climate. The target drought stress in this experiment was ~30% yield reduction. This study identified hybrids that had high-yield in well-watered conditions and minimal yield reduction under drought conditions. There was limited overlap of these “best” hybrids when compared to standard trials in our breeding program, and data from our multi-state GxE project. The differences in the

location/environment and experimental process increases the challenge of producing data comparable to U.S. standard yield trials. Nevertheless, the managed stress-trial did support work on physiological traits, and can be incorporated into prediction models. Incorporation of traits such as anthesis-silking interval, phenology and biomass traits, and physiological traits into prediction models has potential to extend the information and accuracy of those models.

3. Physiological and image-based trait analysis can enrich data from managed stress trials. Component traits, such as anatomical and metabolic traits related to drought tolerance, can support development of breeding and biotech strategies. During the course of this research a new trait, xylem vessel length (identified as the frequency and prominence of perforation plates), was identified to have importance in hydraulic conductance and drought tolerance. For this particular trait, fluid velocity models suggested that the observed variation for perforation plates in maize has strong effects on water transport through xylem conduits, which was supported by in situ measures of water transport across root segments. Maize genotypes with less pronounced perforation plates long metaxylem vessels had greater vulnerability to cavitation events under water stress and displayed lower rates of transpiration under water-stressed conditions. Reduced water use in these genotypes appeared to contribute to a strategy of conserving soil moisture for sustained water capture at later stages of terminal drought stress, as evidenced by their improved plant water status and larger yields at later stages of terminal drought. This discovery adds to the array of root anatomical and architectural traits that have the potential to be used in developing drought-tolerant cultivars.

Data from nutrient and drought stress trials is inherently noisy, with trait heritabilities substantially lower in the stress component of the trial versus the control non-stress. This is a result of inherent microenvironmental variability in the field being exposed due to reduced inputs, as well as systematic variability within plots due to unequal plant-to-plant competition and other considerations that result from the experimental design and process. In this project, we not only measured grain yield but extracted information from ear images using ears collected at specified position within the plots. This information has promise in further refining plant responses in cases where the yield values are similar, but the path to those yield values varied dramatically.

4. Genomic prediction models are most robust when based on composite, large datasets. This is a logical conclusion that was supported in our experiment. Individual trials, and especially stress trials with higher levels of variability, produce genomic prediction models that are limited in their scope of inference. The data from this trial is being evaluated in combination with multi-state multi-location trials using the same germplasm in the context of observed stresses and environmental parameters. The combination of managed-stress trials with known timing and levels of stress together with multi-environment trials can support robust prediction models and facilitate biological interpretation of hybrid performance differences across environments.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;

This study will impact the U.S. food systems by contributing to breeding efforts to develop stress-resilient cultivars. For maize specifically, the study identified specific promising germplasm sources, and provided information and public datasets to support consideration and development of new breeding approaches. More broadly across species, the study provides insights regarding the use of managed-stress trials to develop stress tolerant cultivars. This information is disseminated in refereed publications and is the subject of publications nearing completion.

- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and

Students and post-docs trained during the course of this project will

use the lessons that they have learned as they advance on their career path.

- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Stress tolerant cultivars developed in the future, through breeding processes that incorporate in some form lessons from this research, will support food production nationally and globally using approaches that mitigate climate change and support efficient use of resources.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

The Objectives and Deliverables from the original proposal are below. Commentary on progress toward each Deliverable is embedded within the text.

Objective 1 – Hybrid Evaluation in well-watered versus water-stress conditions to enhance prediction and identify fundamental mechanisms of genomic control

Deliverables

- *Data set of hybrid responses to water-stress based on yield and supporting traits*
A successful trial was conducted each of the three years of the project using germplasm and methodology agreed upon by the funding partners. Partners also included proprietary hybrids in the trial. The data was provided to funders and deposited in the GEMS data resource (gems.umn.edu) for future public access.
- *Genetic analysis of stress-response across hybrid sets*
Phenotypic data was coupled with pre-existing genetic data to conduct genetic mapping and genome-wide association analysis. The results of the analysis are planned to be published in a refereed journal article.

Objective 2 – Interpretation of the differential response of genotypes to drought stress in the context of multi-location trial data

Deliverables

- *Integrated data set of hybrid responses to controlled-stress and dispersed testing environments.*

Phenotypic and environmental data related to this project specifically will be publicly available in a uniform format at GEMS (gems.umn.edu). Genomes to Fields GxE data used for comparisons in this study is available on the resource page

(<https://www.genomes2fields.org/resources/>), after being released with a DOI through Cyverse.

- *List of genomic regions contributing to GxE response, with focus on water limitation and interpretation of the types of genetic control most commonly associated with this differential response.*

This analysis is currently being updated to include the most recent GxE data which included germplasm that overlapped with multiple years of the FFAR trials. An environmental database for the GxE project was developed in part through the FFAR funding to support this analysis.

- *Updated interpretation on the impact of selection for high-yield on the ability of hybrids to respond to stress (i.e. updated analysis based on Gage et al. 2017).*

This interpretation is nearing completion and will be integrated into a publication. As noted earlier, sources of useful alleles for drought tolerance include elite ex-PVP inbreds such as PHN46 in addition to minimally selected inbreds with exotic germplasm such as Mo44. As in all breeding situations, the challenge is incorporating useful alleles for one trait (e.g. stress) when they are often linked to poor alleles for other traits.

Objective 3 – Improving prediction by incorporation of environmental information and by implementation of machine learning.

Deliverables

- *Year 1: Genomic prediction optimized for heterogeneous populations. Parameters of convolutional neural networks tested and optimized.*



- Genomic prediction models have been implemented and used to analyze both GxE and drought stress trials. The results of the latest are included in an extensive internal report that will be the basis for a manuscript. After internal discussions and preliminary analyses, we decided to focus first on parametric models instead of neural networks.
- *Year 2: Genomic prediction for drought related traits of maize hybrids in field-controlled environments implemented.*

Internal analyses are completed, and we are refining the analysis and interpretation as we work toward a publication.

- *Year 3: Validation of prediction models using complete data set.*

To facilitate this goal, we have generated environmental covariate data for GxE trials and from those covariates we identified ~15% of locations in the Midwest with evidence of drought stress during flowering. We are using these 'naturally occurring events' to predict response to drought stress in GxE, and we will incorporate data from the managed-stress trials to predict response in multi-environment trials.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

There were no substantial changes to the project. COVID impacted some aspects of the conduct of the trials. Our analysis has evolved based on our current experiences and knowledge of the team. A data analysis workshop was added to support transfer of knowledge to the funding partners.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)



Our major activities, goals and deliverables, major findings, and conclusions are all outlined in detail in the sections above. A detailed Powerpoint presentation will be provided to the collaborators prior to the final meeting. Manuscript drafts will be provided to the funding partners when submitted. **Information Products**

- 2.7. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Educational videos, scientific publications, date

- 2.8. Please provide a list of citations for the information products produced during the project.

Educational Videos:

<https://www.youtube.com/c/CFStrock>

Publications:

Strock CF, Schneider HM. Root Phenotypes for Improving Nutrient Capture in Low Fertility Environments. In: Goldman I, ed. Plant Breeding Reviews, Vol. 46. Hoboken: John Wiley & Sons, Inc. In press.

Strock CF, Rangarajan H, Black CK, Schafer ED, Lynch JP (2021) Theoretical evidence the root penetration ability interacts with soil compaction regimes to affect nitrate capture. Annals of Botany. In press. <https://doi.org/10.1093/aob/mcab144>

Lynch JP, Mooney SJ, **Strock CF**, Schneider HM (2021) Future Roots for Future Soils. Plant, Cell & Environment, In press . <https://doi.org/10.1111/pce.14213>

Lynch JP, **Strock CF**, Schneider HM, Sidhu JS, Ajmera I, Galindo-Castañeda T, Klein SP, Hanlon MT (2021) Root Anatomy and Soil Resource Capture. Plant and Soil, 466: 21-63. <https://doi.org/10.1007/s11104-021-05010-y>



Schneider HM, **Strock CF**, Hanlon MT, Vanhees DJ, Perkins AC, Ajmera IB, Sidhu JS, Mooney SJ, Brown KM, Lynch JP (2021) Multiseriate cortical sclerenchyma enhance root penetration in compacted soils. Proceedings of the National Academy of Sciences, 118: e2012087118. <https://doi.org/10.1073/pnas.2012087118>

- 2.9. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

The manuscripts cited above involve effort during this project period. Manuscripts directly related to the project are in preparation as we work toward final analysis of the data and, in part, due to the data embargo period.

- 2.10. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

GEMS (gems.umn.edu) is the data repository where data from this project is stored.

- 2.11. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

Not applicable.

- 2.12. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Not applicable.

- 2.13. Beyond depositing information products in a repository, what other activities



have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

Our scientific publications will provide visibility to the data set, and provide useful context and interpretation.

3. Data Management

- 3.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

Three years of phenotypic data, including agronomic, morphological, and yield traits, of controlled experiments; image of harvested ears, weather data, soil moisture and UAV imagery for monitoring trials overall drought response.

Year 1:

- Phenotypic data – traits collected by plot or by row within plot: planting date, harvest date, stand count at germination, stand count before harvest, leaf rolling (three measurements), anthesis date, days to anthesis, anthesis in growing degree days (GDD), silking date, days to silking, silking in GDD, anthesis-silking interval, plant height (rows 1 and 2), ear height (rows 1 and 2), stem diameter (rows 1 and 2), harvest moisture (rows 1 and 2), harvest weight (rows 1 and 2), number of ears (rows 1 and 2), number of barren ears (rows 1 and 2), area of the ten central plants (rows 1 and 2), grain yield (rows 1 and 2).
- Field map: field layout with entries number.
- Weather station data: daily maximum, minimum, and average temperatures (°C), dew point (°C), humidity (%), and wind speed (km/hour), and precipitation (mm).
- Soil moisture sensor data: volumetric water content at depths of 20, 50 and 80 cm from eight soil moisture sensors located in each treatment, replication, and maturity combination.
- Water content data: water content in grams of water in grams of soil, sampled in

three crop development stages, from eight soil samples collected each time.

- Ear images data: Image of ears harvest of the ten central and 4 border plants.
- UAV imagery data: NDVI, NDRE, and OSAVI images collected at three dates during the season.
- Chlorophyll content data: SPAD (soil plant analysis development) measurements were taken on the LH82_PHN46 population on three representative plants per plot in two different dates.

Year 2:

- Phenotypic data – traits collected by plot or by row within plot: planting date, harvest date, stand count at germination, stand count before harvest, leaf rolling (three measurements), anthesis date, days to anthesis, anthesis in growing degree days (GDD), silking date, days to silking, silking in GDD, anthesis-silking interval, plant height (rows 1 and 2), ear height (rows 1 and 2), stem diameter (rows 1 and 2), harvest moisture (rows 1 and 2), harvest weight (rows 1 and 2), number of ears (rows 1 and 2), number of barren ears (plot rows 1 and 2, entire plot and ten central plants), area of the ten central plants (rows 1 and 2), length of ten central plants (m²), distance between rows (cm) grain yield (rows 1 and 2).
- Field map: field layout with entries number.
- Weather station data: daily maximum, minimum, and average temperatures (°F), dew point (°F), humidity (%), and wind speed (mph), accumulated precipitation (in).
- Irrigation data: amount of supplemental water applied through irrigation and irrigation information.
- Ear images data: Image of ears harvest of the ten central and 4 border plants.
- UAV imagery data: NDVI, NDRE, and OSAVI images collected at three dates during the season.

Year 3:

- Phenotypic data – traits collected by plot or by row within plot: planting date, harvest date, stand count at germination, stand count before harvest, leaf rolling (three measurements), anthesis date, days to anthesis, anthesis in growing degree days (GDD), silking date, days to silking, silking in GDD, anthesis-silking interval,

plant height (rows 1 and 2), ear height (rows 1 and 2), stem diameter (rows 1 and 2), harvest moisture (rows 1 and 2), harvest weight (rows 1 and 2), number of ears (rows 1 and 2), number of barren ears (rows 1 and 2), area of the ten central plants (rows 1 and 2), grain yield (rows 1 and 2), irrigation information.

- Field map: field layout with entries number.
- Weather station data: daily maximum, minimum, and average temperatures.
- Soil moisture sensor data: volumetric water content at depths of 30, 60, 90 cm from nine soil moisture sensors located in each treatment, and maturity combination.
- Water content data: water content in grams of water in grams of soil, sampled in three crop development stages, 13 soil samples collected total.
- Irrigation data: amount of supplemental water applied through irrigation and irrigation information.
- Ear images data: Image of ears harvest of the ten central and 4 border plants.
- UAV imagery data: NDVI, NDRE, and OSAVI images collected at three dates during the season.

- 3.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

All the datasets are part of the three years of evaluation and complimentary to achieve the goals proposed.

- 3.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Metadata information was provided for each year in readme files.

Year 1: "Read Me.docx"

Year 2: "Read Me 2019-2020.docx"

Year 3: "Read Me Year 3.docx"

Year 1 Content:

DRT_U_Wisconsin

- Range: Refers to a single horizontal pass of the field. Values are 2-15 as ranges 1 and 16 are boarder.
- Row: Refers to a single vertical pass of the field. Values consist only of odd numbers because all plots are two row plots (the even row 4 is the pair to the odd row 3, 6 with 5, etc.).
- Replication: The rep that the entry belongs to.
- Entry: Six digit identification system used by Tuniche. The first digit is a field designation and is always 1. The second digit refers to the replication within the treatment. The third digit refers to the maturity status. The next three digits are unique plot identifiers and are concurrent 001-461 for the early maturity block and 001-389 for the late maturity block.
 - Well-watered treatment rep 1 early block = 111xxx.
 - Well-watered treatment rep 2 early block 121xxx.
 - Well-watered treatment rep 1 late block 112xxx.
 - Well-watered treatment rep 2 late block 122xxx.
 - Drought treatment rep 1 early block 131xxx.
 - Drought treatment rep 2 early block 141xxx.
 - Drought treatment rep 1 late block 132xxx.
 - Drought treatment rep 2 late block 142xxx.
- Treatment: WW = well watered and DRT = drought. The 1's and 2's refer to the replication.
- Leaf Rolling 1, 2, and 3: The drought treatment was scored for leaf rolling on a 1-9 scale.
 - 1: Corn without expression of water stress, no apparent difference in leaf changes during the day.
 - 3: Maize with the 2 upper leaves showing a slight degree of the edge of the leaf curved inwards. None of the leaf margins overlap. It begins to detect a slight grayish color and / or less bright on the leaves.
 - 5: 50% of the leaves of the plant show a significant degree of curl. There is an easily detectable color change (gray or discolored leaves).
 - 7: All the sheets are rolled, the margins of the sheets touch each other and overlap. Very significant change in color (grayish and less bright leaves).

burned.

- 9: Maximum leaf curl. Several tips of the leaves will begin to feel

3

5

7

9



- Date 50% Silks, Pollen, and ASI: The date at which 50% of the plants in a plot have silked out, started to shed pollen, and the anthesis silking interval.
- Plant Height, Ear Height, and Stem Diameter Row 1 and 2: Tuniche measured a single plant in row 1 and row 2 for each entry and reported them separately along with the average of the two values. The following table presents some entries that have been discarded or

should be approached with caution. Potential and significant outliers were identified using Grubbs Test. Complimentary graphs can be found in the spreadsheet titled “DRT_U_Wisconsin_Graphs.” Any entry with values listed as “suppressed” were removed by Tuniche due to low confidence in the quality of the plot or measurement.

Element	Plot	Entries	Notes
Height Row 1			Value and potential outlier
Height Row 2			Height of 115cm less than ear height of 120cm. Row 1 values 335cm (PH) and 135cm (EH)
Height Row 1 vs Row 2		122103, 121255, 111050, 121193, 122225, 121116, 112312	Difference between row measurements > or equal to 100 cm
Height Row 1		12311	Height values, respectively. Both outliers.
Height Row 1 vs Row 2		12311	Differs as above. Difference between rows equals 80cm for both entries.
Stem Diameter Row 1		122343, 122033, 122283, 122333, 121284	> 35mm
Stem Diameter Row 2			> 35mm
Stem Diameter Row 2		122309, 122031, 122285, 122352, 121244, 121310, 122375, 122358, 112182, 122361, 122357	> 35mm
Stem Diameter Row 1 vs Row 2		121116, 122358, 122361, 111040, 112182, 122094, 121159, 122044, 122033, 112078, 122092, 121284	Difference between rows > 10mm
Stem Diameter Average		121244, 122343, 122357	> 35mm
Height Row 2		141111, 142314	Heights of 28 and 25 cm, respectively
Height Row 1 vs Row 2		142242, 131440, 131410, 131396, 132176	Difference between rows > or equal to 100cm
Height Row 1		132006	Best values and potential outliers.
Height Row 1		132269	Best values and potential outliers.
Height Row 1 vs Row 2		132221, 131392, 132269	Entries listed from row 1. Difference between rows is > 90cm for all entries besides 13221 (difference of 60cm)
Stem Diameter Row 1		141441, 142368	> 35mm
Stem Diameter Row 2		141217	> 35mm
Stem Diameter Row 1 vs Row 2		141053, 141185, 141395	Difference between rows equal to or > 10mm
Stem Diameter Average		141217	> 35mm

- Harvest moisture and weight: Measurements (percent moisture and weight in kg) were taken on 10 ears within the row as well as 4 border ears (2 on each end of the row) in order to track the boarder effect. The following table presents some entries that are notable or should be approached with caution. Potential and significant outliers were identified using Grubbs Test. Complimentary graphs can be found in the spreadsheet titled “DRT_U_Wisconsin_Graphs.” Any value listed as “mv” is a missing value.

Measurement	Location	Entries	Notes
Moisture 10 Ears Row 1		121110	Moistures > 30%. Significant outliers. 6 More entries with moistures > 25%
Moisture 10 Ears Row 2			Moistures > 25%
Moisture 10 Ears Row 1 vs Row 2		122222	Difference between rows equal to or > 10%
Moisture 10 Ears Average		121110	High values from row 1. Entry 122065 significant outlier.
Moisture 4 Ears Row 1			1 outlier. 6 other entries with moistures > 25%
Moisture 4 Ears Row 2			1 of 32.4%, significant outlier. 7 entries with moistures > 25%
Moisture 4 Row 1 vs Row 2			Difference > 10%. Also large difference between rows in 10 Ear measurements
Moisture 4 Ears Average		(one of which is 122065)	Moistures > 25%
Weight 10 Ears Row 1		122018	Highest values and significant outliers.
Weight 10 Ears Row 1		122308	Highest values. Weakly significant outliers.
Weight 10 Ears Row 2		5 other entries	1 notable value and significant outlier / All entries under 0.5kg.
Weight 10 Ears Row 1 vs Row 2			Differences > 1kg
Weight 10 Ears Average			1 value, potential outlier.
Weight 4 Ears Row 1		122039, 122199	Highest values all under 0.2kg.
Weight 4 Ears Row 1		122287, 112252, 111375, 121213, 122296, 112009	12 entries with some spread from the rest of the data.
Weight 4 Ears Row 2			1 notable entries under 0.2kg and 9 highest entries above 1.2kg.
Weight 4 Ears Row 1 vs Row 2		122009, 112252	Difference between rows of approximately 1kg
Weight 4 Ears Average		12328, 112296, 122287	Average weights.
Moisture 10 Ears Row 1			1 value and potential outlier.
Moisture 10 Ears Row 2		141302, 141100	High values and potential outliers.

Moisture 10 Ears Row 1 vs Row 2				e between rows > 10%
Moisture 10 Ears Average		.41302		values and potential outliers.
Moisture 4 Ears Row 1		.42036, 141302		values and significant or potential outliers.
Moisture 4 Ears Row 2		.42164, 141100		values and significant or potential outliers.
Moisture 4 Ears Average		.41302, 142036		values and significant or potential outliers.
Weight 10 Ears Row 1				weight < 0.1 kg
Weight 10 Ears Row 2				weight < 0.1kg
Weight 10 Ears Row 1 vs Row 2		131454, 132188, 132046, 141461, 141062, 132277, 141349, 131087, 142076		e between rows > 1.0kg
Weight 10 Ears Average		29 Entries		weight < 0.1kg
Weight 4 Ears Row 1				weight < 0.1kg. Entry 142162 has weight of 0kg
Weight 4 Ears Row 2				weight < 0.1kg
Weight 4 Ears Average		.41431		weight < 0.1kg

- Stand count, harvest ears count, and barren count: Stand count at harvest, the number of ears harvested from each row, and the number of barren plants and or ears in a row.
- m2 10 ears: The total area (in square meters) covered by the 10 plants from which the ears were harvested. This value was used to calculate the yield in kilograms per hectare.

SPAD_data_pop_lh82_phn46

SPAD measurements were taken on the LH82_PHN46 population. This population was selected due to the large amount of variation present in leaf rolling scores and total yield in previous experiments. Leaf rolling scores are also included in this spreadsheet for convenience.

- SPAD_1, SPAD_2, and SPAD_3: SPAD measurements were taken on 3 separate plants in a plot on the 28th and 29th of January 2019.
- 2SPAD_1, 2SPAD_2, and 2SPAD_3: repeat measurements were taken again on the 5th of February, 2019.

Weather Station Data

This spreadsheet contains weather data collected by Tuniche from two research stations. The data consists of the maximum, minimum, and average temperatures (°C), dew point (°C), humidity (%), and wind speed (km/hour). The maximum, minimum, and total amount of precipitation (mm) is provided for research station two. Only the total amount of precipitation was available at research station one. The third sheet consists of graphs that were produced by our Tuniche collaborators.

Soil Moisture Sensor Data

There were 8 soil moisture sensors spread throughout the field such that one was located in each treatment, rep, and maturity status combination. Measurements of volumetric water content were taken at depths of 20cm, 50cm, and 80cm. Sheet titles are interpreted as well-watered (WW) or drought (DRT), Late (L) or Early (E) maturity status, and either rep 1 or 2. Once again, the last sheet contains graphs produced by our Tuniche collaborators.

Water content_soil samples_gravimetric method

Soil samples were taken in each of the 8 treatment/maturity/repetition combinations and sent to a lab for analysis. This was done on three separate dates (December 28th, 2018, January 24th, 2019, and February 15th, 2019). The first three sheets contain the lab results and the fourth sheet (consolidated) consists of some graphs produced by our Tuniche collaborators.

Ear Images Folder

This folder contains photographs of the ears harvested from the two replications of the drought treatment. The file names are interpreted as such, using the first image from rep 1 as an example. 111001_Hilera1_Rep_1_Bloque_2_Hilera_3 → Entry 111001_Row1_Rep_1_Block_2_Row_3 This image is thus of the first row (of two) for entry 111001 from the first rep, found in the second range and third row of the field. See field map for clarification. There will be a second image associated with entry 111001 for the ears harvested from the second row of the plot.

NDVI Folder

- Drone pictures with index
 - Three files for the three dates that drone images were taken (January 5th, January 23rd, and February 19th, 2019)
 - Contains images of the fields with NDVI, NDRE, and OSAVI measurements. The fourth image shows the total surface area of each block.
 - This output came from a Spanish company and therefore all the text is in Spanish.
- Comparison data NDVI
 - This file was put together by our Tuniche collaborators and attempts to provide a quantitative measurement of the drone images.
 - A range of NDVI values was selected and the total area of the image that has NDVI values within that range was calculated.
 - This spreadsheet may or may not be useful, but it has been provided as an example of another way to view the data.

The following folders contain the raw data files for NDVI, NDRE, and OSAVI images. Tuniche outsources all drone imaging to another company and this is what they produced when asked for the “raw data.” Many of the file names are in Spanish.

- SHP_IndicesValoresABS
 - Zipped folder containing 72 items



- There are 8 file types each of the three measurements (NDVI, NDRE, and OSAVI) on each of the three dates. For example, NDRE_050119 refers to the NDRE data collected on the 5th of January, 2019.
 - DBF, PRJ, SBN, SBX, SHP, QIX, XML, and SHX files
 - DBF can be opened in Microsoft Excel
 - SHP files (shapefiles) can be opened in the program AgroSIG. I believe/assume that this is the program used to generate the SHP files.
 - SBN, SBX, and SHX files are associated with shapefiles
 - The program ESRI ArcGIS and its associated add-ins may be useful for the viewing of shapefiles. There are likely other options as well.
- KMZ_IndicesdeVegetacion (Vegetation Indices)
 - 3 zipped folders for each of the three dates images were taken.
 - Each folder has 3 KMZ files, one for NDRE, NDVI, and lastly OSAVI
 - KMZ files can be viewed in Google Earth. Contains geographic annotations.
- AgroSIG_Tuniche
 - 2 zipped folders for each of the three dates images were taken. The folder titled “AgroSIG_Tuniche_Graneros_230119” contains folders for both of the January dates. Each folder contains the following items.
 - IMAGEN Multispectral folder
 - TIF images and associated TFW and OVR files.
 - RASTER INDICES folder
 - TIF images and associated TFW and OVR files for NDRE, NDVI, and OSAVI
 - SHP folder containing SHP and associated files
 - An SLS and SPR File are also present

DRT_U_Wisconsin_Graphs

Complimentary graphs for the “DRT_U_Wisconsin” spreadsheet. These basic graphs depict the raw data and do not present any novel information and are only meant to help guide more in-depth analysis. There are four sections.

- Sections 1 and 2: the data was separated by treatment and scatterplots were made. Section one depicts all applicable scatterplots of the well-watered treatment and section two depicts the drought treatment. The x-axis is labeled as “Observation” for all graphs. Rather than plotting entry number on the x-axis, each entry was assigned a single digit observation code and that was plotted instead. There is no significance to the order in which the observation code was assigned and any apparent spatial trend in a given

scatterplot is meaningless. All outliers were tested via Grubbs Test. Any mention of significance without providing a p-value means the p-value was well under 0.005.

- Section 3: Pearson's correlations were calculated between the two rows of an entry as well as between the average of the two replications. The lowest correlations have been highlighted.
- Section 4: Heatmaps of the raw data as well as the residuals calculated from a basic model fit. These are provided to help guide more sophisticated analysis of potential spatial variation across the field. The raw data plots are always presented before the corresponding residual plot.

2018_2019 Field Map

Field map consisting of all entries and plot numbers. Rep 2 is stacked on top of rep1. The well-watered treatment is shaded in green and the drought treatment is shaded in red. The early block will be a darker shade of green or red than the late block. Each plot number is listed twice because the field map shows each individual row rather than each plot.

SOP

There are two standard operating procedures documents. The one titled SOP_Tuniche is the original provided by Tuniche and is in Spanish. The one titled SOP_Translated is simply the original document after being translated with Google Translate. The translation is not perfect, but it is good enough to get the point across.

The SOP outlines the agronomic practices Tuniche followed and their techniques for collecting the data found in the other spreadsheets. The term block is used interchangeably with the term range to refer to a horizontal pass down the field.

Year 2 Content:

***Points to keep in mind**

- All plots are two row plots. Data was collected and recorded on each row individually. References to individual rows will be made. Averages of the two rows are provided in order to produce a single value for a plot. This is reflected on the field map, where each plot entry number is listed twice.
- Harvest data was collected separately for border plants and middle plants within a row. This was done to isolate the border effect that individual rows experience and observe this affect under drought conditions. For each row, there will be measurements done on the 4 border plants/ears. These border plants are the 2 plants at the front and back of a row. For

the middle of the row, 10 representative plants were tagged. Any phenotypic measurements came from the border plants or the 10 middle row plants and will be specified.

Drought Trial 2019-2020 Phenotypic Data

- Treatment: WW for “well-watered” and DRT for “drought” treatments.
- Rep: the replication that the entry belongs to
- Genotype: Pedigree of the entry
- Group: The experimental design for the trial consists of multiple populations. The group designation places a genotype into its respective population. There are five groups:
 - Check - experimental checks that remain constant across years of the experiment
 - PHW65 - PHW65 populations that were crossed with LH195
 - W10004 - W10004 lines crossed with PHK76
 - PSU_Hybrids - various hybrids requested to be included in the trial by collaborators
 - PSU_Inbreds – various inbreds requested to be included in the trial by collaborators
- Maturity: Genotypes were considered to have either an early (e) or late (l) maturity status based on previously collected data on flowering time.
- Range: Refers to a single horizontal pass of the field. Values are 1-13. Border ranges on either end of the field ARE NOT included. See field map.
- Row: Refers to a single vertical pass of the field. Values consist only of odd numbers because all plots are two row plots (row 2 is the even pair to odd row 1, 4 is paired with row 3, etc.). See field map.
- ENTRY: 6 digit numeric code given to each plot. The breakdown is as follows.

Group	Rep 1 Well Watered	Rep 2 Well watered	Rep 1 Drought	Rep 2 Drought
PHW65	111001-111163	121001-121163	131001-131163	141001-141163
	112001-112159	122001-122159	132001-132159	142001-142159
W10004 Early	151001-151223	161001-161223	171001-171223	181001-181223
W10004 Late	152001-152205	162001-162205	172001-172205	182001-182205
PSU Inbreds	301001-301006	302001-302006	303001-303006	304001-304006
PSU Hybrids	412001-412030	422001-422030	432001-432030	442001-442030

*There is another group with entries starting with a 2 (2***** series). These were genotypes from private sector collaborators. Their data is not included in this file, but 2***** series numbers will be reflected on the field map.

- Planting Day: The day in November of 2019 that the entry was planted (either the 28th, 29th, or 30th).
- Stand count Average (Row1-Row2) Germination: Stand counts of each row were taken post germination. In order to provide a count for a single plot, the two rows were averaged.
- Leaf Rolling 1, 2, and 3: Plants in the drought treatment was scored for leaf rolling on a 1-9 scale.
 - 1: Corn without expression of water stress, no apparent difference in leaf changes during the day.
 - 3: Maize with the 2 upper leaves showing a slight degree of the edge of the leaf curved inwards. None of the leaf margins overlap. It begins to detect a slight grayish color and / or less bright on the leaves.
 - 5: 50% of the leaves of the plant show a significant degree of curl. There is an easily detectable color change (gray or discolored leaves).
 - 7: All the sheets are rolled, the margins of the sheets touch each other and overlap. Very significant change in color (grayish and less bright leaves).
 - 9: Maximum leaf curl. Several tips of the leaves will begin to feel

burned.

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- Date 50% Pollen: The date at which 50% of the plants in a plot had begun shedding pollen.
- Date 50% Silk: The date at which 50% of the plants in a plot had silked out.
- ASI (Difference): Anthesis silking interval.
- Plant Height, Ear Height, and Stem Diameter (Row 1 and Row 2): Since plots are two row plots, each entry had a single measurement taken from each row. A single plant from the middle of row 1 was selected and plant height (measured to the top of the tassel in cm), ear height (primary ear's node in cm), and stem diameter (mm) were recorded. The same process was repeated for row 2 of a plot. The average of the two plants was calculated.
- Anthesis days, silking days, anthesis GDD, and silking GDD: days from planting until anthesis and silking are listed. Growing degree days for each are also provided.
- Stand count (before harvest) (Row 1 and Row 2): Stand counts were taken again before harvest. This time stand count values are recorded for each row of a plot individually.
- Total ears (before harvest) (Row 1 and Row 2): The number of plants that had viable ears was counted in each row of a plot.
- # Barrens (in all plants in Row) (Row 1 and Row 2): The number of barren plants was counted for each row. This includes border plants and the 10 selected inner plants.
- Barren percentage (Row 1 and Row 2): The percentage of plants that were barren in a row is also provided.



- # Barrens (in the 10 plants selected in Row) (Row 1 and Row 2): The number of barren plants that were located within the 10 plants selected for harvest was recorded for a given row.
- Surface (mt²), distance between rows (cm), and length of 10 plants (cm) (Row 1 and Row 2): These columns describe the surface area found within a row, the distance between rows, and the distance between the first and last plant in the 10 selected middle plants. These numbers are used for calculating yield.
- Harvest day: The day an entry was harvested. Falls within the range of April 16th – May 7th, 2020.
- Harvest moisture 10 ears, 4 border ears (Row 1 and Row 2): Percent moisture at harvest was calculated for the 10 middle ears and for the 4 border ears in each row. Averages are also provided.
- Harvest weight (kg) 10 ears, 4 border ears (Row 1 and Row 2): Ears were harvested and weights were taken for the 10 middle ears and for the 4 border ears in each row. Averages are also provided.
- Comments (Row 1 and Row 2): Any necessary comments regarding harvest moisture and weight calculations are provided here. These will provide insight on why no data is listed for some entries.
- Yield (Kg/Ha) (Row 1 and Row 2): Yield was calculated in Kg/Ha using the plot size, harvest moisture, and harvest weight measurements. This was done for each row of a plot separately. Averages are also provided.
- Yield (bu/ac) (Row 1 and Row 2): Yield is also provided in bushels per acre.

Field Map 2019-2020

This file is self-explanatory. B = Border (green shaded cells are hybrid border; yellow shaded cells are inbred border). The number in the colored cells correspond to the entry numbers found in the phenotypic data file. Ranges are numbered 1-13 to the left of each rep/treatment block and rows are numbered 1-288 below each replication block. Each entry is listed twice because plots are two row plots.

Irrigation Data 2019-2020

This file describes the irrigation protocol for the trial. The file is broken into three sheets.
Sheet One: Timing and Amount



- This sheet gives the area of the field that each irrigation valve flowed into, the date, length of time that the irrigation was turned on, and the volume of water per hectare in cubic meters, cubic millimeters, and cubic inches. There is an additional comments column where Tuniche explains why they added additional irrigations to the schedule.
- There were 6 valves that were used to irrigate the entire trial. The portion of the trial that a valve flowed into and the total area that a valve fed are found in the first three columns.

Sheet Two: Summary

- A small table that shows the total volume of water that each rep/treatment received throughout the growing season.

Sheet Three: Valves 19-20

- A version of the field map that is color coded to show which portions of the field each valve flowed into.

Weather Data 2019-2020

The high, low, and average temperature, dew point, humidity, and windspeed was recorded throughout the growing season. The total amount of accumulated precipitation is also listed.

Drone Images Folder

Drone imaging was done three times throughout the growing season to collect NDVI, NDRE, and OSAVI images. The dates of the flights were January 17th, February 7th, and February 28th, 2020. There are three file types for each flight.

1. Images_Tuniche_DATE: These PDF files contain the images produced by the drone imaging company that Tuniche hires to do the flights. Due to this, the text is in Spanish. The additional fourth image titled "Superficie por Bloque" is used to calculate the surface area of each section of the field.
2. Summary_Tuniche_DATE: These excel files (also in Spanish) provide a breakdown of each treatment/replication block. Ranges/Bins of values were created for each index (NDVI, NDRE, OSAVI). The total area of the field (in Hectares) that had an index value within the range for a given bin was calculated. The average value for the total area is provided (column D). Data for each index is on a separate sheet within the file.
3. KMZ files for each date are also provided.

Ear Images 2019-2020

This data set includes the images taken of the harvested ears from the 2019-2020 drought trial in

Graneros, Chile. The file directory is broken down by replication, treatment, and ultimately range of the field.

Main folder

➔ Rep 2

➔ Rep 1

-> DRT

-> WW

-> WW REP1 RANGE 1

-> Images

-> WW REP1 RANGE 2

-> Images

Etc. for all 13 ranges of the field in each treatment/replication.

There were four images taken per plot. Plots are two row plots, with two images coming from each row. The images are of the 4 border plant ears and 10 middle plant ears that were harvested from each row. File names indicate the plot number, which row of the plot the ears came from, and if the image is of the border or middle plant ears. For example, the image named 151001_Row_1_4_Ears is of the 4 border ears harvested from row 1 of plot 151001. There will be three other images of this plot, with names 151001_Row_1_10_Ears, 151001_Row_2_4_Ears, and 151001_Row_2_10_Ears.

Labels were included in each image that state the plot number and the physical row number that it came from. Plot numbers (ie entry numbers) are listed as the "location" on the corn board and the row number is listed as the "product." (Note: these are not the Row 1 and Row 2 designations for the plot but the physical row number from the field. See field map included in the "Drought 2019-2020 Data" dataset).

There are a handful of images where the name of the image does not match the location and product lines included in the image. These were mistakenly labeled by Tuniche when the images were taken, but the image file name should always be correct. For concerns, please see the excel document titled "File with comments of the pictures." Filter the comments column to remove the blank spaces and all that remains will be the images that were mislabeled. The spreadsheet includes what the actual plot number was, what it was wrongly labeled as, and a comment saying that the name was correctly edited.

* If a file is found whose name does not match the label in the image and IS NOT listed in this spreadsheet as being fixed, please contact Ryan Alpers (ralpers@wisc.edu) and he will sort it out for you.

Year 3 Content:

*Points to keep in mind

- All plots are two row plots. Data was collected and recorded on each row individually. References to individual rows will be made. Averages of the two rows are provided in order to produce a single value for a plot. This is reflected on the field map, where each plot entry number is listed twice.
- Harvest data was collected separately for border plants and middle plants within a row. This was done to isolate the border effect that individual rows experience and observe this affect under drought conditions. For each row, there will be measurements done on the 4 border plants/ears. These border plants are the 2 plants at the front and back of a row. For the middle of the row, 10 representative plants were tagged. Any phenotypic measurements came from the border plants or the 10 middle row plants and will be specified.
- This is the raw data. The only data cleanup that has been done was the removal of nonsensical values. It is suggested that you filter the phenotypic data by the comments column in order to identify entries with missing values before proceeding with analysis.

Drought Year Three Phenotypic Data

- Sector - A consequence of the split field design. Refers to which field the entry was located in (see field map).
- Range - Refers to a single horizontal pass of the field. Values are 2-15. Border ranges on either end of the field in ranges 1 and 16.
- Row - Refers to a single vertical pass of the field. Values consist only of odd numbers because all plots are two row plots (row 2 is the even pair to odd row 1, 4 is paired with row 3, etc.). See field map.
- Entry - 6 digit numeric code given to each plot. The breakdown is as follows.

Group	Rep 1 Well Watered	Rep 2 Well watered	Rep 1 Drought	Rep 2 Drought	Rep 3 Drought
Main Trial Early Maturity	141001-141317	151001-151318	111001-111440	121001-121440	131001-131440
Main Trial Late Maturity	142318-142475	152319-152475	112441-112675	122441-122675	132441-132675

PSU Inbreds	340001- 340006	NA	310001- 310006	320001- 320006	330001- 330006
PSU Hybrids	440001- 440030	NA	410001- 410030	420001- 420030	430001- 430030

*There is another group with entries starting with a 2 (2***** series). These were genotypes from private sector collaborators. Their data is not included in this file, but 2***** series numbers will be reflected on the field map.

- Population - The experimental design consists of multiple populations. These can be roughly broken down into the following categories:
- Treatment - DRT for drought and WW for well-watered
- Maturity - Pedigrees were considered to have either an early or late maturity status based on previously collected flowering time data.
- Material Group - The experimental design for the trial consists of multiple populations. The group designation places a genotype into its respective population. There are five groups:
 - Check - experimental checks that remain constant across years of the experiment
 - PHW65 - PHW65 populations that were crossed with LH195
 - W10004 - W10004 lines crossed with PHK76
 - W606S – populations derived from W606S crossed with either PHK76, PHN46, or PHP02 that were then crossed with PHT69
 - LH82 – populations derived from LH82 crossed with either PHN46 or PHP02 that were then crossed with PHT69
 - LH185 – populations derived from LH185 crossed with either PHP02, PHK76, or PHN46 that were then crossed with PHT69
 - 3IIH6 – populations derived from 3IIH6 crossed with either PHP02 or PHK76 that were then crossed with PHT69
 - PHN46_3IIH6 – population crossed with PHT69
 - PHK76_LH82 – population crossed with PHT69
 - Mo44_PHW65 – population crossed with LH195
 - PHN11_PHW65 – population crossed with LH195
 - PSU_Inbreds – various inbreds requested to be included in the trial by collaborators
- Stand count Average (Row1-Row2) Germination - Stand counts of each row were taken post germination. In order to provide a count for a single plot, the two rows were averaged.



- Leaf Rolling 1, 2, 3 - Plants in the drought treatment was scored for leaf rolling on a 1-9 scale.
 - 1: Corn without expression of water stress, no apparent difference in leaf changes during the day.
 - 3: Maize with the 2 upper leaves showing a slight degree of the edge of the leaf curved inwards. None of the leaf margins overlap. It begins to detect a slight grayish color and / or less bright on the leaves.
 - 5: 50% of the leaves of the plant show a significant degree of curl. There is an easily detectable color change (gray or discolored leaves).
 - 7: All the sheets are rolled, the margins of the sheets touch each other and overlap. Very significant change in color (grayish and less bright leaves)
 - 9: Maximum leaf curl. Several tips of the leaves will begin to feel burned.

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- Stop Irrigation - The day irrigation was turned off
- Supplemental irrigation - irrigation turned back on for a short period to prevent crop loss
- Stop supplemental irrigation - the day the supplemental irrigation was turned off

Note – Graneros, Chile received rain on the 29th of January

- Date 50% Silk - The date at which 50% of the plants in a plot had silked out.
- Date 50% Pollen - The date at which 50% of the plants in a plot had begun shedding pollen.
- ASI - Anthesis silking interval.
- Plant Height, Ear Height, and Stem Diameter (Row 1 and Row 2) - Since plots are two row plots, each entry had a single measurement taken from each row. A single plant from the middle of row 1 was selected and plant height (measured to the tassel in cm), ear height (primary ear's node in cm), and stem diameter (mm) were recorded. The same process was repeated for row 2 of a plot. The average of the two plants was calculated.
- Stand count (Row 1 and Row 2) - Stand counts were taken again before harvest. This time stand count values are recorded for each row of a plot individually.
- Harvest ears (Row 1 and Row 2) - The total number of plants that had viable ears was counted in each row of a plot.
- Barren Count (Row 1 and Row 2) - The number of barren plants that were located within the 10 plants selected for harvest was recorded for a given row.

- Harvest moisture 10 ears, 4 border ears (Row 1 and Row 2) - Percent moisture at harvest was calculated for the 10 middle ears and for the 4 border ears in each row. Averages are also provided.
- Harvest weight (kg) 10 ears, 4 border ears (Row 1 and Row 2) - Ears were harvested and weights were taken for the 10 middle ears and for the 4 border ears in each row. Averages are also provided.
- m² 10 ears (Row 1 and Row 2) - the surface area, in meters squared, that the 10 selected plants inhabited. Used in calculating yield.
- Yield (Kg/Ha) (Row 1 and Row 2) - Yield was calculated in Kg/Ha using the plot size, harvest moisture, and harvest weight measurements. This was done for each row of a plot separately. Averages are also provided.
- Yield (bu/ac) (Row 1 and Row 2) - Yield is also provided in bushels per acre.

Sensor Measurements and Lab Data

- Irrigation tab – describes the actual amount of water applied to each treatment via irrigation throughout the season.
 - Broken down by maturity (C + L = companies + late, E = early, I = inbreds)
- Flowmeter tab – data collected from flowmeters to monitor the performance of select irrigation valves.
- Moisture samples tab – lab results of the analysis of soil sample water content. Samples were collected in each treatment / replication.
- Moisture Sensors tab – data collected from the in-field moisture sensors.
- GDU tab – data describing the growing degree units each day of the year
 - Dates are formatted day/month/year
 - Columns B and C are in Celsius, E and F are in Fahrenheit

Field Map

Mostly self-explanatory. Since all plots are two row plots, each entry number is listed twice. Border rows are represented by a B. Irrigation valves for each section of the field are also labeled (for example V 3 Sur or V 10 Norte).

Ear Images (only located in gems database)

Ear images are not included on the pen-drive due to the total size of the files. They can be found in the COTF Corn Drought Team on the GEMs platform (<https://gems.agroinformatics.org>). For assistance accessing GEMs, please contact Ryan Alpers at ralpers@wisc.edu.

There were four images taken per plot, two images per row. The images are of the 4 border plant ears and 10 middle plant ears that were harvested from each row. File names indicate the plot

number, which row of the plot the ears came from, and if the image is of the border or middle plant ears. For example, the image named 111001_R1_NS_1524 is of the 10 mid-row ears harvested from row 1 of plot 111001. There will be three other images of this plot, with names 111001_R1_B_1523, 111001_R2_B_1525, and 111001_R2_NS_1526. NS indicates that the ears are of the 10 mid-row ears and B indicates that they are the border ears. The numbers following either NS or B are a part of the generic file name generated by the camera.

There was a planting error that affected 10 plots. There are no images for entries 111331, 111332, 122500, 122501, 151093, 151094, 220124, 220125, 240209, 240210. (Note - the 2***** series are company related plots and are not found in the main spreadsheet)

Drone Data

Folder containing all files related to the three drone flights performed on January 19th, January 25th, and March 11th, 2021. Flights on each date are broken into two folders (DRTC1 and DRTC2). The C's are short for "Corte," which is reflected in some other folder and file names. Corte translates to "cut" in English and refers to the "section" designation on the field map. In other words, each field of the split field design was flown separately and the data is presented in their own folders / files. C1 is for field section 1 and C2 is for field section 2.

The data found in this folder is the raw data and consists of a multitude of file types and results typical of a drone flight.



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RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: ____Shawn Kaeppler_____

PI Signature: *Shawn Kaeppler*

Date: November 30, 2022

Authorized Signing Official (ASO) Name: Angie Johnson

ASO Signature: _____ Date: _____